

Week 9

Marketing and Data Science



松尾・岩澤研究室
MATSUO-IWASAWA LAB UTOKYO

01

Data Science in the World of Business

- What is a **Business**?
- ★ How does **Data Science connect to the Business World**?
- Key Example: Introduction to Data Science in **Marketing**

Final Assignment Advice

02

Hands-on Marketing Analysis (15min Video. Designed for Self-study)

- Demonstration using Jupyter Notebook
- Data exploration, decile analysis, RFM analysis, etc.
- Recommendation engine, causal inference, etc.

03

The Next Steps

- ★ Data Science as YOUR Business: What makes a good proposal?
- Theoretical to **Practical Knowledge**
- **Starting your own business**

Final Assignment Advice

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Data Science in the World of Business

01

What is a Business?

What is a Business



The world runs around businesses. But what are they?

Making Money? Making Profit?

What is a Business



The world runs around businesses. But what are they?

Making Money? Making Profit?

Building an Organization?

What is a Business



The world runs around businesses. But what are they?

Making Money? Making Profit?

Building an Organization?

Generating Growth?

What is a Business



The world runs around businesses. But what are they?

Making Money? Making Profit?

Building an Organization?

Generating Growth?

Becoming Famous?

What is a Business



The world runs around businesses. But what are they?

Making Money? Making Profit?

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Becoming Famous?

Becoming Really Really Rich?

What is a Business



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Isaacson, W. (2023). *Elon Musk*. Simon & Schuster.

What is a Business



The world runs around businesses. But what are they?

Making Money? Making Profit?

Building an Organization?

Generating Growth?

Becoming Famous?

Becoming Really Really Rich?

What is a Business



The definition

business

noun

UK  /'biz.nɪs/ US  /'biz.nɪs/

business *noun* (SELLING)

Add to word list 

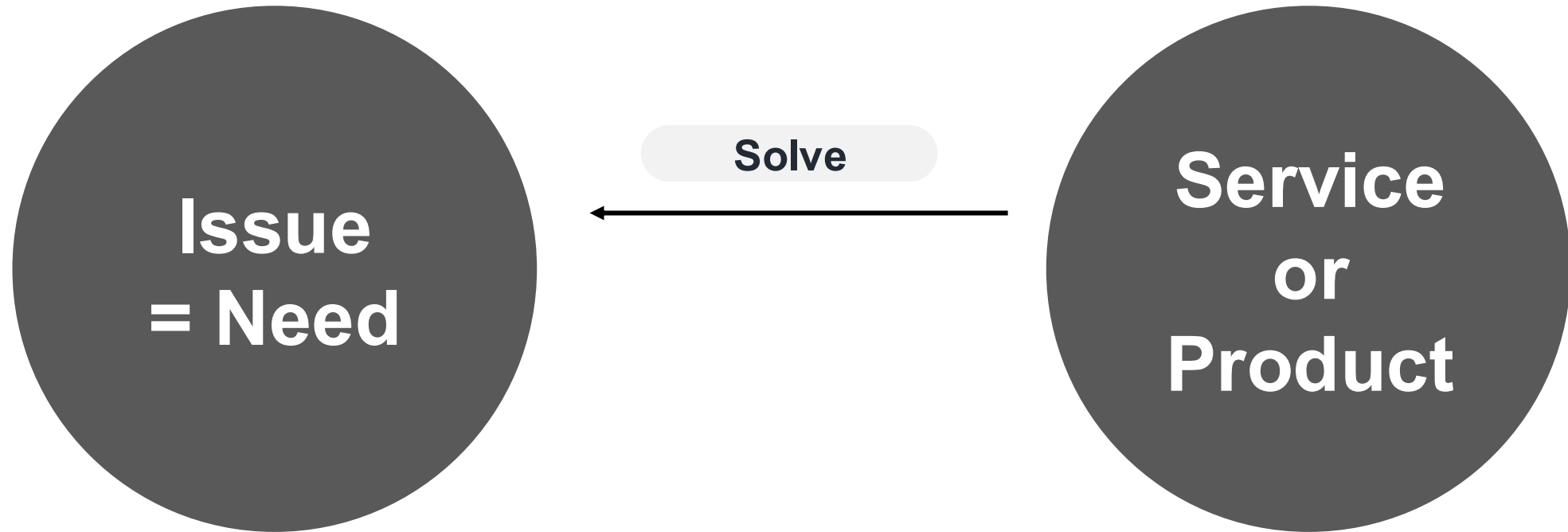
A1 [U]

the activity of buying and selling goods and services:

Business: The act of solving an issue and getting paid



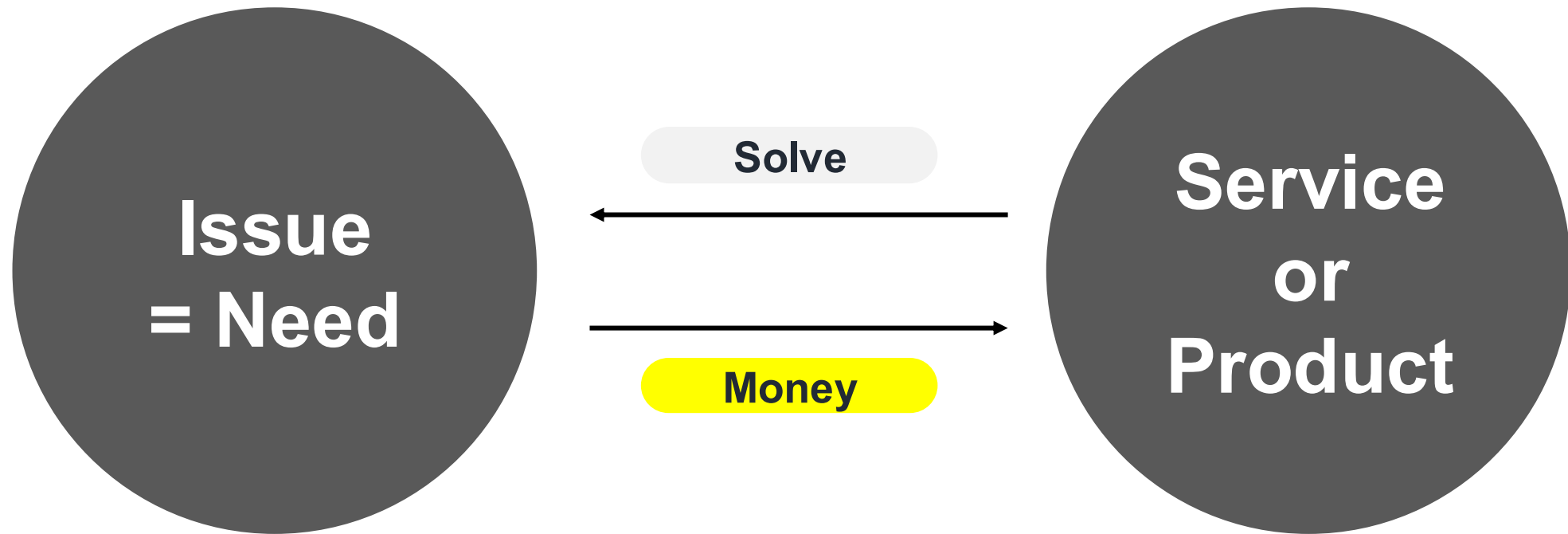
There's always an issue or a need, then there comes the business



Business: The act of solving an issue and getting paid



There's always an issue or a need, then there comes the business



The size of your issue directly resonates the size of your business

Final Assignment Advice

What makes a good issue?

01

It is fundamental, and therefore has a large impact



The monthly view of this streaming channel is going down. Who's our core repeater? What's their motivation?



The monthly view of this streaming channel is going down. How much more do I need to invest in production?

02

It comes with a deep / suggestive hypothesis



Maybe there is a correlation between streaming views and the season?



I think the content is out of date. We might need new casts, new content?

03

You know it can be answered



What do our monthly repeat customers want, and how can we satisfy them effectively?



What is the future of streaming services?

A set of checklists that you can use



Please ask yourself if your problem / issue statement is solid using the following checklist.

Balancing all three is the difficult part (01 makes it larger, but 03 narrows it down)

Final Assignment Advice

01 It is fundamental, and therefore has a large impact

- ✓ If we answer this, what decision does it change? What do we do differently tomorrow?
- ✓ What's the implication of solving it — does it redirect strategy, or just tweak a detail?
- ✓ How much money moves if we get it right? (market size, revenue at stake, cost saved)
- ✓ How many stakeholders are affected, and how big are they? (users, customers, internal teams)
- ✓ Is this a root cause, or a symptom of something deeper?
- ✓ If we don't answer it, what breaks? (cf. Companies hate risks more than opportunities)

02 It comes with a deep / suggestive hypothesis

- ✓ Do I have a specific stance, or just a vague "let's investigate"?
- ✓ What common belief or assumption does this challenge?
- ✓ Have I found a pattern, relationship, grouping, or rule that reframes the problem?
- ✓ Is the hypothesis falsifiable — could data prove me wrong?

03 You know it can be answered

- ✓ Can existing methods/technology resolve this?
- ✓ Is the data available, or obtainable?
- ✓ Can I define what a clear "yes / no" answer looks like?
- ✓ Is the scope narrow enough to finish, but broad enough to matter?

Quick Check of Understanding - Are these good issues?



Let's look into examples in the Marketing scene



01 Monthly Active User Retention / Turnover

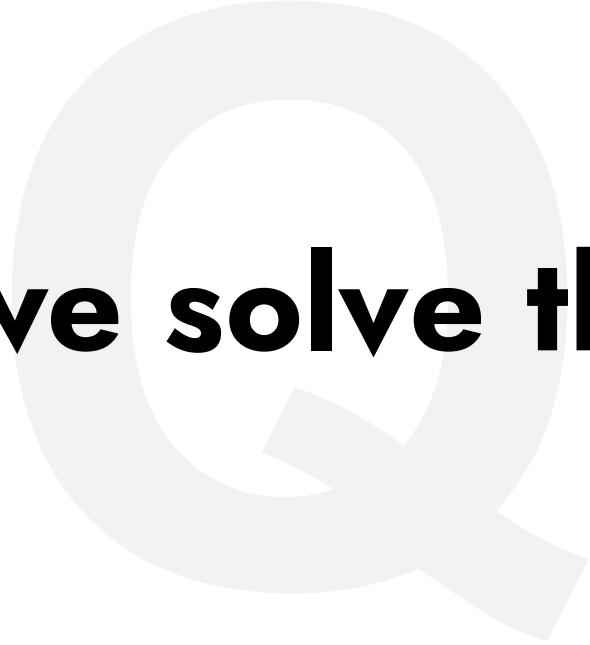
- ✓ ☒ Strong: "Which subscriber segment drives most of our cancellations, and what viewing pattern predicts their churn one month out?" — fundamental (churn is the core revenue threat), hypothesis-driven (it's concentrated in a segment with a leading signal), answerable (watch logs + subscription data).
- ✓ ☒ Weak: "How do we make all our users happier so nobody cancels?" — not fundamental, no testable stance, unbounded scope.

02 Personal Activities / Engagement

- ✓ ☒ Strong: "At which point in a session do viewers stop browsing without pressing play, and is it driven by too many choices rather than weak catalog?" — pinpoints a decision (fix the browse step), carries a counter-conventional hypothesis (choice overload > content gap), testable with session clickstream.
- ✓ ☒ Weak: "Is our recommendation system good?" — vague, no hypothesis, no clear answer.

03 Investment to Content

- ✓ ☒ Strong: "Do titles released with a structured 4-week promo window reach their viewership peak faster than those without, and by how much?" — measurable, comparative, changes release strategy if true.
- ✓ ☒ Weak: "What is the future of streaming?" — interesting but unanswerable, no decision attached.



How do we solve the issue?

Solving these issues is where data science comes in



Looking into the data is key to each issue



01 Monthly Active User Retention / Turnover

- ✓ "Which subscriber segment drives most of our cancellations, and what viewing pattern predicts their churn one month out?"

Set of Data you want to look into: subscription records (signup, plan, cancel date), per-user watch-time & session trend over 30–60 days, login gaps, genres viewed, segment attributes (channel, device, region).

02 Personal Activities / Engagement

- ✓ "At which point in a session do viewers stop browsing without pressing play, and is it driven by too many choices rather than weak catalog?"

Set of Data you want to look into: session clickstream (browse → play or exit), browse time before abandoning, number of titles shown vs. scrolled past, play vs. abandon sessions.

03 Investment to Content

- ✓ "Do titles released with a structured 4-week promo window reach their viewership peak faster than those without, and by how much?"

Set of Data you want to look into: per-title release & promo metadata, daily viewership curves (days-to-peak, peak size), promo spend, non-promo titles as control



It doesn't start from the solution.
It starts from the issue

Example: Application in the Marketing Field



Why Marketing?



Marketing is one of the ideal starting point for real-world application

1. Clear, Measurable Goals (KPIs & KGIs):

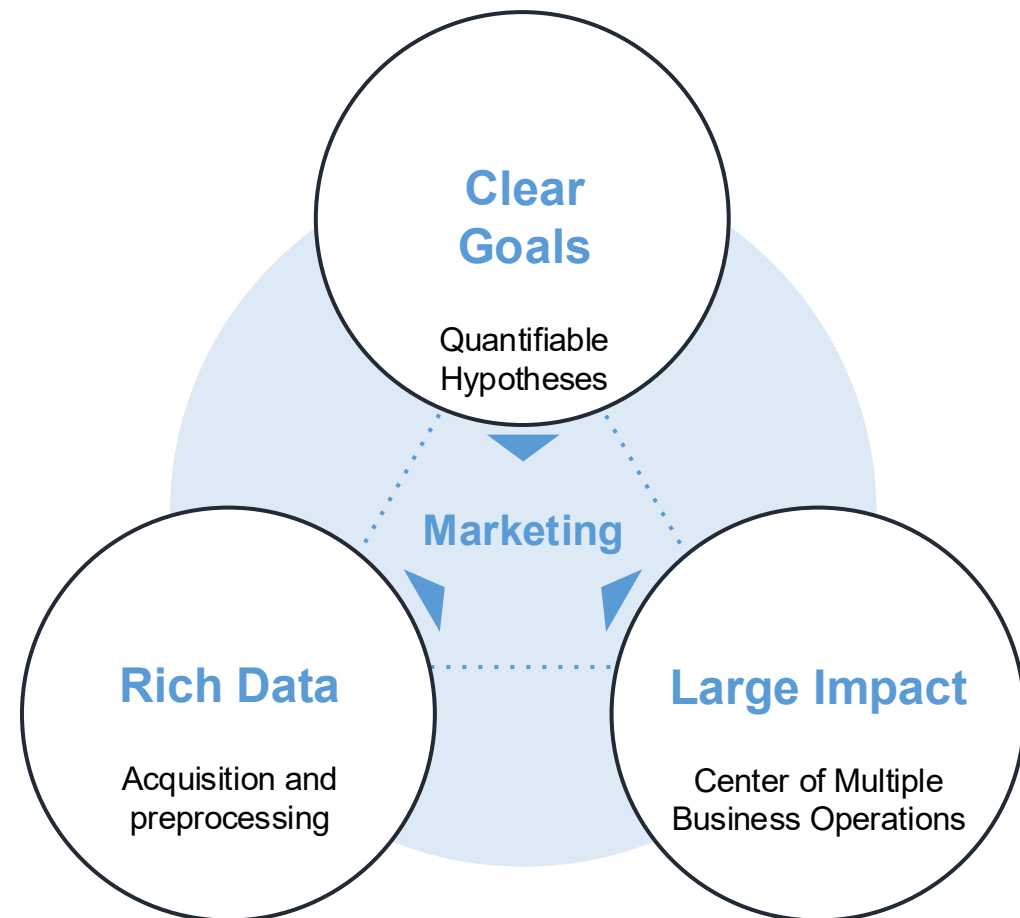
- **Well-defined, measurable goals** (increasing sales, customer retention, reducing churn, and improving customer engagement etc.)
- These goals naturally translate into **quantifiable hypotheses**

2. Rich Availability of Data:

- Inherently produces **vast quantities of data**, such as:
 - Behavioral, Transactional, Demographic data, etc.
- An ideal domain for practicing data acquisition and preprocessing skills.

3. Center of many business operations:

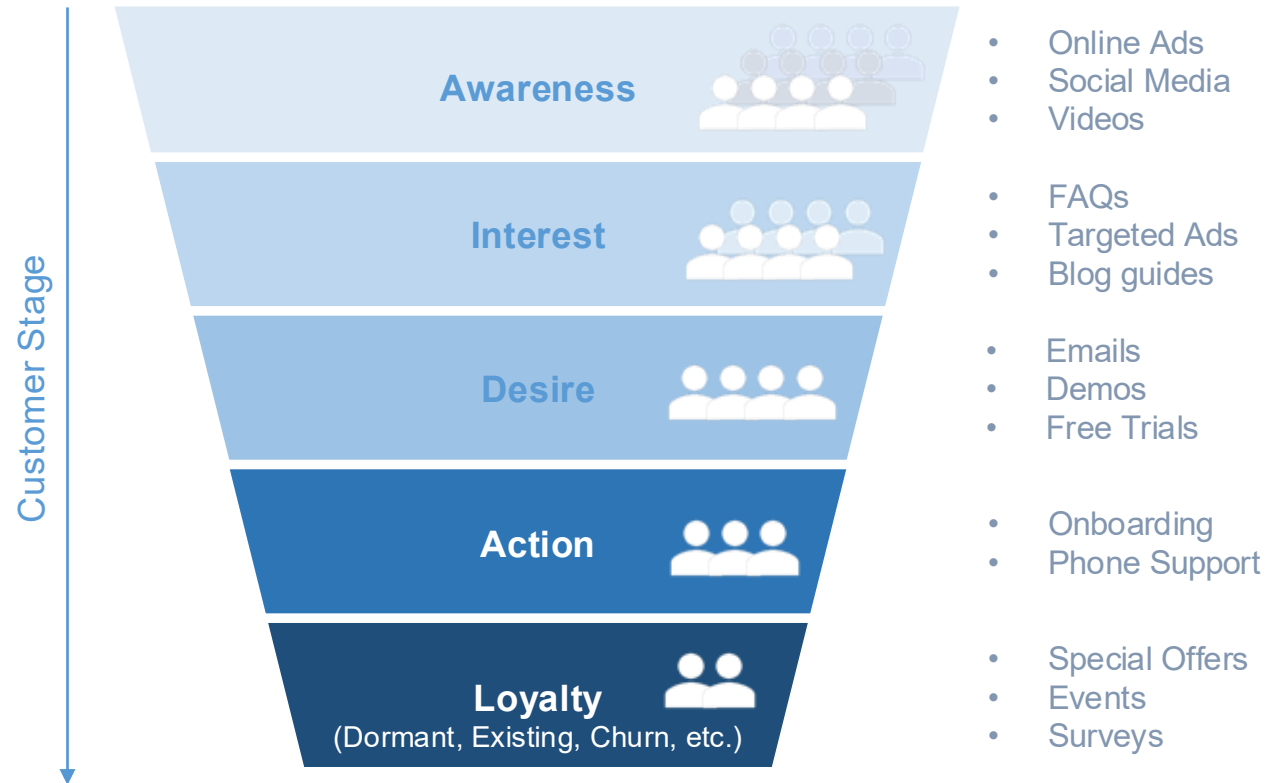
- Often serves as a hub connecting various departments such as R&D, Sales, Customer Support, and Supply Chain Management
- **Hence holding a large impact in business scenes**



Fundamentals of Marketing



Marketing is the process of creating strategies and systems to sell products or services **profitably and efficiently**.



Key Marketing Questions:

- **Which stage** is this specific customer in?
- How can we efficiently **attract new customers**?
- What encourages existing customers to **spend more**?
- How do we **re-engage dormant customers**?
- Which stage has an issue in **conversion rate**?
- Can we **predict and prevent customer churn**?

Customer Stage Varies Among Customers and Requires Different Approaches.
⇒ Data is Used to **Identify, Personalize, and Evaluate** Marketing Strategies

Data Usage & Strategies in Familiar Services

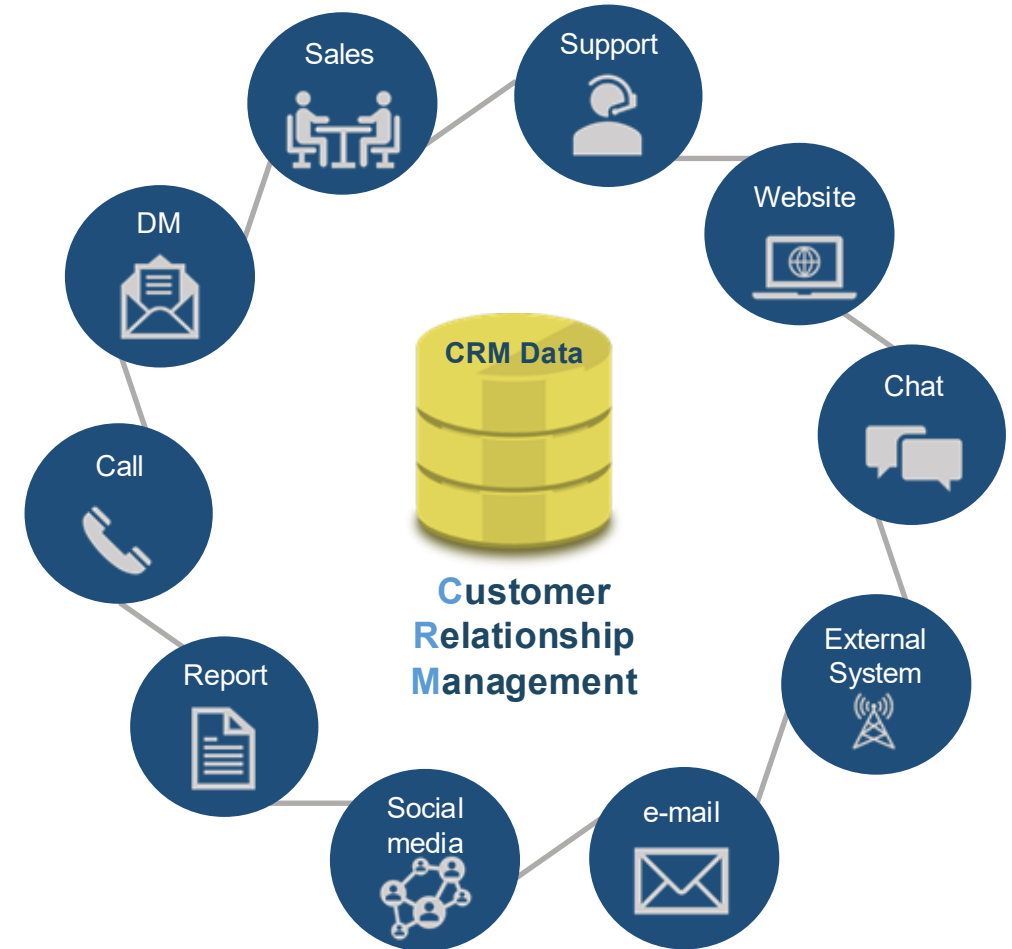


How Marketing Strategies are Implemented in our Daily Lives



Data that is Being Used

- **Browsing and Interaction Data:** User clicks, session duration, content viewed, user engagement.
- **Transactional and Purchase Data:** Purchase history, transaction frequency, order value.
- **Demographic and Profile Data:** Age, location, preferences, and customer segment information.
- **Behavioral and Temporal Data:** Frequency, recency, patterns in customer behavior over time.
- **Feedback and Sentiment Data:** Customer reviews, ratings, and survey responses.



Data Usage & Strategies in Familiar Services



How Marketing Strategies are Implemented in our Daily Lives



What We Experience

- **Netflix:** Personalized movie recommendations based on watch history.
- **Amazon:** Targeted offers based on browsing and purchase history.
- **Airbnb:** Tailored travel recommendations and experiences based on user profiles and previous bookings.
- **Starbucks:** Personalized rewards and promotions based on customer transaction data.

Customers who viewed items in your browsing history also viewed

Collaborative Filtering

Related to items you viewed

Content-based Filtering

The screenshot displays two sections of Amazon's recommendation engine. The top section, titled 'Customers who viewed items in your browsing history also viewed', features a carousel of book covers with the label 'Collaborative Filtering'. The bottom section, titled 'Related to items you viewed', features a carousel of electronic products like a USB adapter, headphones, and earbuds with the label 'Content-based Filtering'. Each item in the carousel includes its title, rating, price, and a 'Get it as soon as' date.

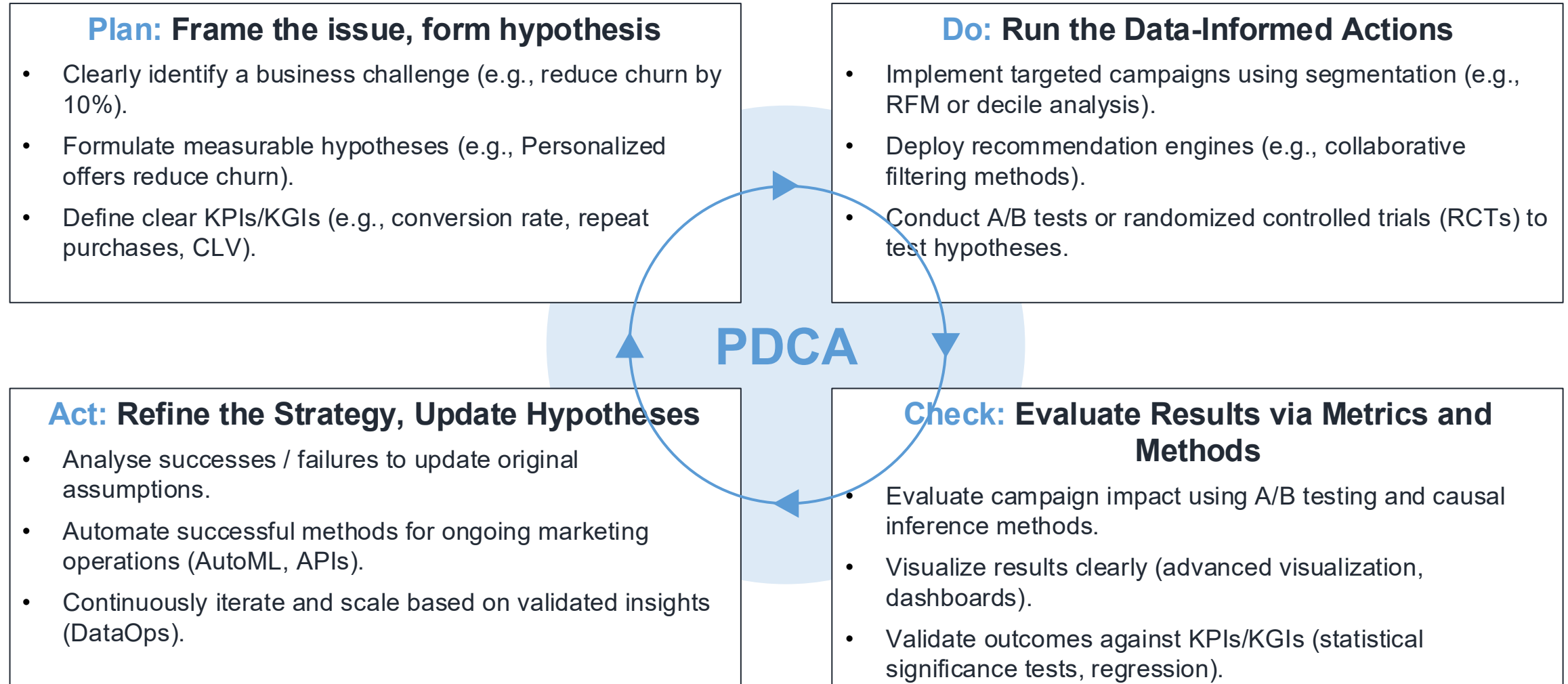
Intro and Example of PDCA Cycle



The PDCA Cycle of Data Science in Business World



Hypothesis Driven Analysis, Data-Informed Action, and Constructive Evaluation through the PDCA Cycle



The PDCA Cycle of Data Science in Business World



Plan: Frame the issue, form hypothesis

- Clearly identify a business challenge (e.g., reduce churn by 10%).
- Formulate measurable hypotheses (e.g., Personalized offers reduce churn).
- Define clear KPIs/KGIs (e.g., conversion rate, repeat purchases, CLV).

Do: Run the Data-Informed Actions

- Implement targeted campaigns using segmentation (e.g., RFM or decile analysis).
- Deploy recommendation engines (e.g., collaborative filtering methods).
- Conduct A/B tests or randomized controlled trials (RCTs) to test hypotheses.

PDCA

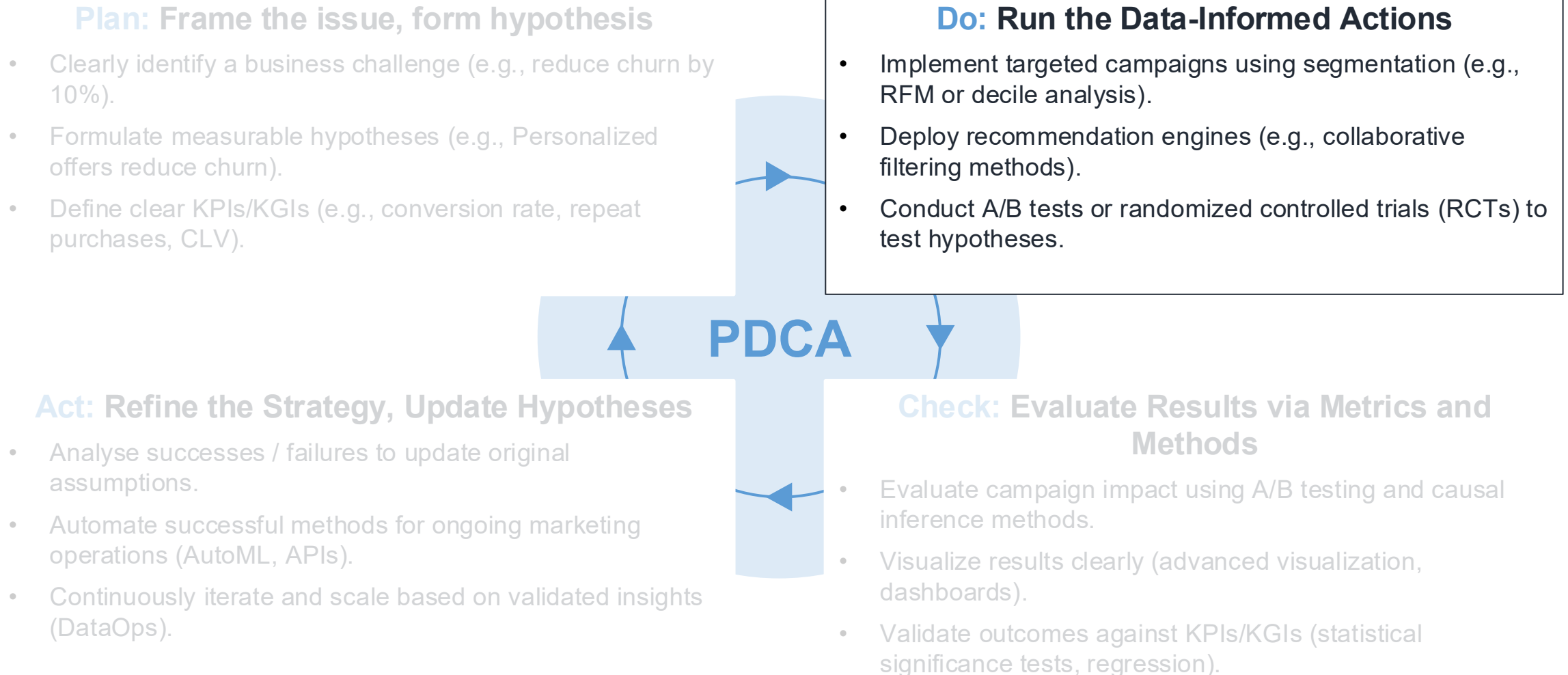
Act: Refine the Strategy, Update Hypotheses

- Analyse successes / failures to update original assumptions.
- Automate successful methods for ongoing marketing operations (AutoML, APIs).
- Continuously iterate and scale based on validated insights (DataOps).

Check: Evaluate Results via Metrics and Methods

- Evaluate campaign impact using A/B testing and causal inference methods.
- Visualize results clearly (advanced visualization, dashboards).
- Validate outcomes against KPIs/KGIs (statistical significance tests, regression).

The PDCA Cycle of Data Science in Business World



The PDCA Cycle of Data Science in Business World

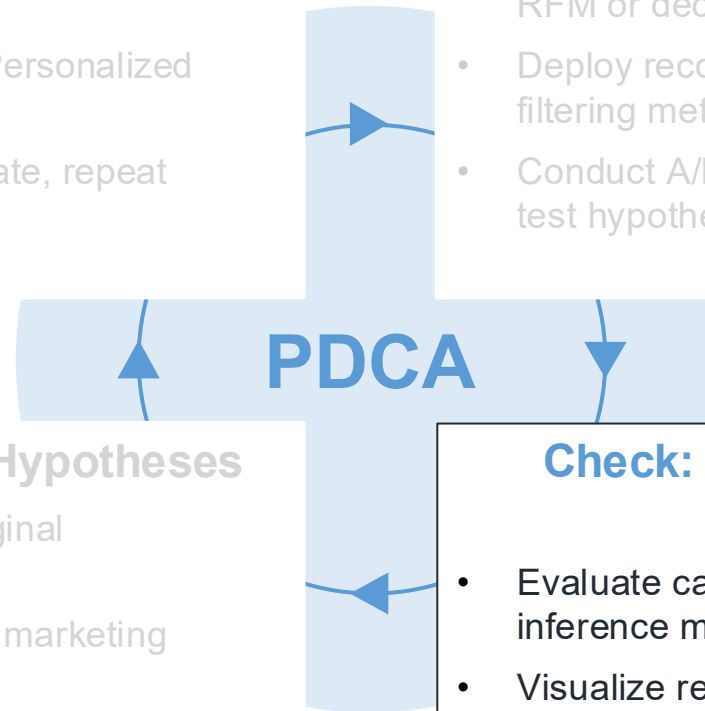


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Final Assignment Advice



Hands-on Marketing Analysis

(Designed for Self-study)

02

Jupyter Notebook contents for this lecture

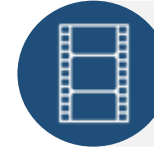


Introduction to Consumer Analysis and CRM in Marketing

- Marketing strategy
- Data exploration
- Decile analysis
- RFM analysis
- Separation analysis
- Factorization

The screenshot shows the Table of Contents for the notebook "11 Introduction to Consumer Analysis and CRM in Marketing". The left sidebar lists the following sections: 11.1 The Background and Overall Picture of this Chapter, 11.1.1 Consumer Behavior Analysis and CRM, 11.1.2 Three Key Points in Consumer Behavior Data Analysis, 11.1.3 Identify Issues, 11.1.4 There is no silver bullet in data analysis, 11.1.5 The Overall Structure of This Chapter, 11.2 Marketing Strategy and Data Analysis, 11.2.1 Marketing Strategy in Business, 11.2.2 What is KPI?, and 11.2.3 Data Strategy and IT. The main content area shows the "Chapter Opener" for 11.1, which includes a paragraph about exploring consumer analysis and CRM within marketing strategy, and a list of two bullet points: "To understand the basic approaches for analyzing CRM data in marketing and to implement representative methods (such as decile analysis, RFM analysis, and churn analysis) using Python for analysis." and "To be able to identify three key points in consumer data analysis."

Introduction



Advanced Consumer Analysis and CRM in Marketing

- Recommendation
- Causal inference
- Automation through advanced visualization, APIs, etc.

The screenshot shows the Table of Contents for the notebook "11 Advanced Consumer Analysis and CRM in Marketing". The left sidebar lists the following sections: 11.1 Background and Overview to This Chapter, 11.1.1 Customer Proposals, Campaign Evaluation, and Automation, 11.1.2 Libraries Mainly Used in This Chapter, 11.2 Recommendation, 11.2.1 What you will learn in this chapter, 11.2.2 Collaborative Filtering, Practice Question, Answer, 11.2.3 Content-Based Filtering, 11.2.4 Hybrid Methods, Deep Learning, and Reinforcement Learning for Recommendation, 11.2.5 Recommendation Evaluation Methods, and 11.3 Causal Inference. The main content area shows the "Introduction" for 11.1, which includes a paragraph about learning a slightly more advanced approach to marketing analysis, and a list of two bullet points: "To understand the basic approaches for analyzing CRM data in marketing and to implement representative methods (such as decile analysis, RFM analysis, and churn analysis) using Python for analysis." and "To be able to identify three key points in consumer data analysis."

Advanced

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Final Assignment Advice



The Next Steps: Looking ahead to the Final Assignment

03

Most companies can't solve their own data-science problems — **so solving them pays**

Final Assignment Advice

This is why taking GCI has such a huge impact on your career value.

Companies' issues = your opportunity



No in-house talent

They lack people with data science skills.



Messy, unusable data

Data exists but isn't in a usable shape.



Tech moving too fast

AI / DS changes faster than they can keep up.

You solve it → you get paid

As a data scientist, the problems companies can't solve are your market. That's how data science becomes a business — yours.

...or they solve it themselves.

Which also means it's a job opportunity for you

Netflix: recommendation models drive most of what people watch — directly cutting churn. The few who build it in-house turn data science into their edge.

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Expectation of Final Assignment: A Data-driven Proposal

What makes a Good Proposal?



The backbone of a business proposal. But as aforementioned, **it all starts from a good solid issue**

Final Assignment Advice



Sharp issue

A specific, sizeable problem worth solving. A quick background research helps



Clear hypothesis

A testable claim about how to solve it



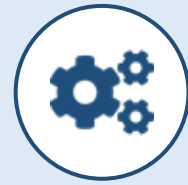
Testable with data

You can verify or verified it with evidence / experiments



Quantified impact

Show the expected size of the prize (KPI, ROI)



Feasibility

Realistic given data, budget, and time, also, Why You?

Bottom line: a good proposal connects an impactful problem to data — and shows the money. You get the margin



Starting Your Own...?

01



Continue learning

- Participate in Kaggle competitions and actively study successful case studies.
- Enroll in advanced courses on deep learning and emerging technologies (e.g., no-code tools, cloud services).
- Expand your international knowledge by strengthening language skills, particularly English.

02



Experimental Development

- Develop software or analytical systems to build practical experience.
- Experiment with creating web applications, social media integrations, drone and robotics operations, automation solutions, video editing projects, game development, or financial trading platforms.

03



Go into the field

- Engage in part-time or internship roles involving data analysis, system development, etc.
- Obtain hands-on project experience to solidify your technical and practical skills.
- Continuously acquire new competencies and knowledge directly applicable to your professional tasks.

Many Companies Are Struggling



01

Lack of Data Strategy & In-house Capability

02

Complex and Bizarre Databases

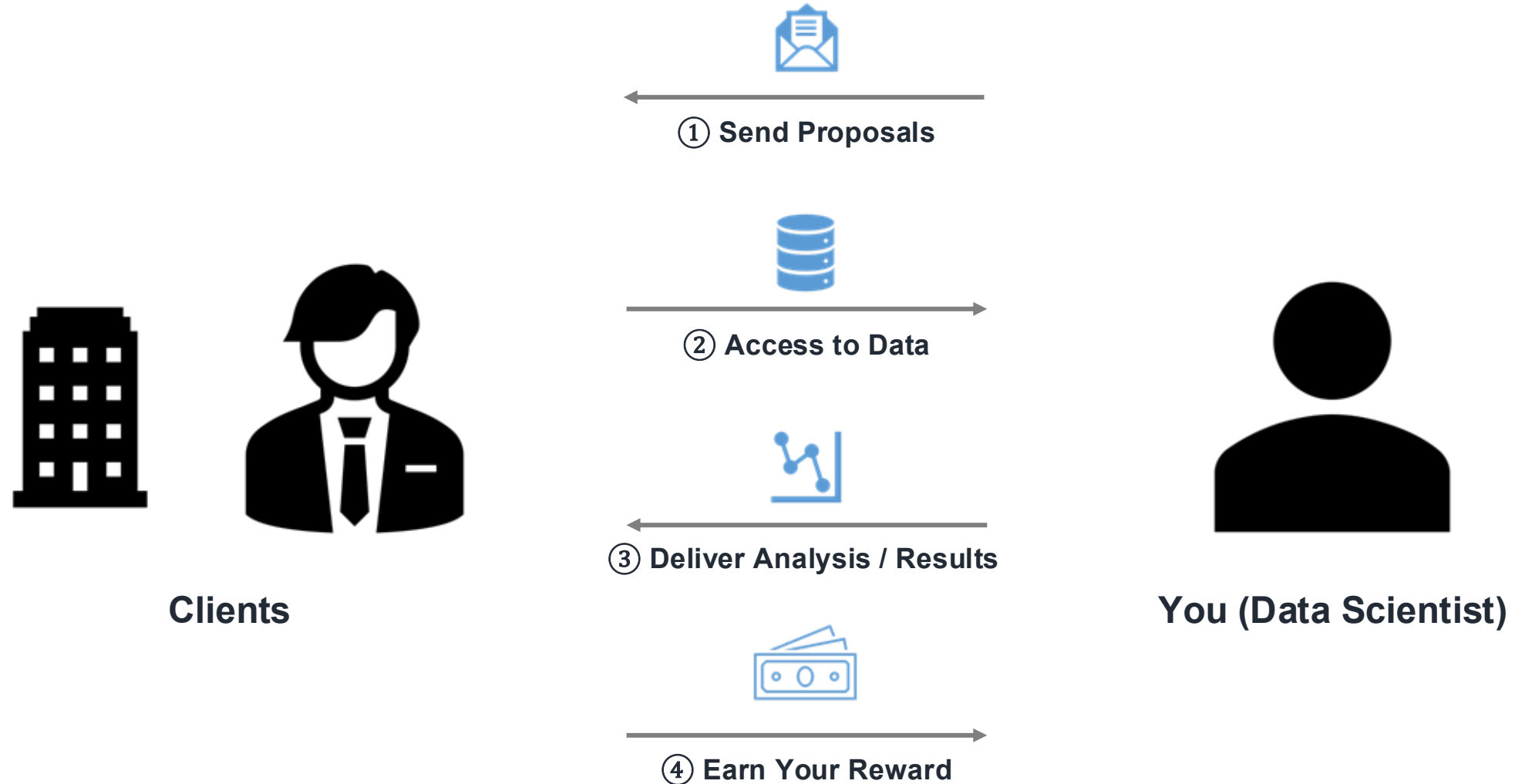
03

Radical Transformation in Technology, Particularly in AI and Data Science

The **ample time** and the **proximity to cutting-edge technology** are invaluable.
With the right skills, you can contribute meaningful value from **day ONE**

What we are doing here at GCI is applicable in the market

You can start small — either on your own or with a team.



Once You Get the Hang of It, Found One



Matsuo Lab Supports Entrepreneurial Journeys



① Get Ready

- Take GCI and related courses
- Get sufficient knowledge and skills around data science / AI etc.



② Build a Team

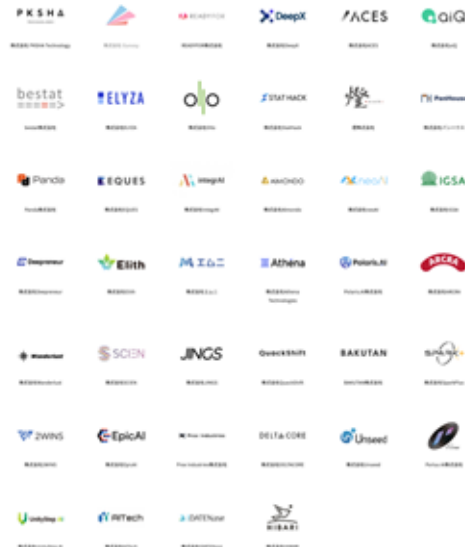
- Find Co-founders
- Find first few clients
- Deliver and Build Reputation



③ Found your Company

- Build an Organization
- Continue the Cycle of Sales, Delivery and Recruitment
- Build your Own Product

List of Matsuo Lab Startup® 40+ companies (as of May, 2026)



IPO 2 companies

PKSHA
TECHNOLOGY

Sep, 2017



Apr, 2015

M&A 3 companies

ELYZA
KDDI

Mar, 2024

STAT HACK
KAMINASHI

Jun, 2024

Awarded Forbes JAPAN 30 Under 30

2022



2024



01

Data Science in the World of Business

- ✓ Business is the act of [solving an issue and getting paid](#)
- ✓ A good issues is [Fundamental, is Hypothesis Driven, and Solvable](#)
- ✓ Data Science come into play when we want [to solve these issues](#)

Final Assignment Advice

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Hands-on Marketing Analysis (Designed for Self-study)

- ✓ Demonstration using Jupyter Notebook
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- ✓ Recommendation engine, causal inference, etc.

03

The Next Steps

- ✓ Data Science can be your OWN Business, or Value
- ✓ A good proposal [starts with a solid issue](#), backed by a hypothesis, data analysis, goal measures, and feasibility
- ✓ Taking GCI is a quick head start to starting your own business – [Let's find our own good issue!](#)

Final Assignment Advice





